Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* Global Collaborative R&D Program
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA Global Collaborative R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA Global Collaborative R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Resilient Display Technology: Advanced Dye-Based Polarizing Films for Automotive and Outdoor Applications			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical, Electronics		Chemicals	
	Medium Category	Small Category	Medium Category	Small Category
	Display	Display Components and Materials	Fine Chemicals	Dyes / Pigments and Intermediates
Concept of Proposed Technology	**Background and unmet need** • Most commercial polarizers for displays still rely on iodine-doped, uniaxially stretched poly(vinyl alcohol) (PVA) because they provide high polarization performance and high contrast [1,2]. • Such Iodine-PVA polarizers can be vulnerable to harsh environments (high temperature, high humidity, or/and UV), where iodine desorption/sublimation and microstructural relaxation can lead to light leakage and performance drift [3,4], eg. commercial iodine-based polarizers degrade significantly under 85°C/85%RH conditions, losing polarization efficiency (bleaching) within 200 hours. • On the other hand, dichroic dye-based polarizers can be a promising alternative when higher heat/humidity resistance is required [4]. **Core challenge** • Achieving both (i) high polarization efficiency (PE ≥ 98–99%) at practical single-piece transmittance (Tv ≈ 38–45%) and (ii) long-term environmental durability remains non-trivial. • PE is governed by chromophore optical anisotropy (dichroic ratio/transition-dipole alignment), dye concentration, film thickness, and processing history, and may show non-monotonic behavior with			

	draw ratio [5]. • Recent studies report improved durability (e.g., via reactive dyes) and even high-PE dye polarizers under demanding conditions, but these advances are often case-specific. Generalizable structure–property–durability design rules and manufacturable pathways are still limited [4]. **Proposed core technology** This project will establish molecular and structural design principles that decouple PE from durability in dye-based polarizers, linking dye structure, dye–PVA interaction/fixation, and alignment stabilization to the optical and aging performance. The end goal is a manufacturable dye-based polarizer prototype meeting PE ≥ 98% (stretch target 99%) at Tv of 38–45% with durability suitable for automotive/outdoor/military displays.
Rationale for Joint R&D Project	This is not an incremental process-tuning project. The key technical bottleneck is a mechanism-level understanding of the PE–durability trade-off in dye-based polarizers and the translation of that understanding into actionable design rules. University of Toronto (UT) will contribute molecular/structural analysis and modeling capabilities (polymer physics, polarized spectroscopy, and structure-property modeling) required to: • quantify the theoretical and practical limits of PE as a function of dye optical anisotropy, concentration, and film structure • identify how dye–PVA interactions (e.g., hydrogen bonding vs. covalent fixation/crosslinking) govern alignment stability and degradation under heat/humidity. [4-6] The Korean SME partner will provide industrial polarizer fabrication capabilities (dyeing, stretching, lamination) and reliability evaluation, enabling rapid prototype iteration and validation under application-relevant conditions. The TIPA Designated (Pre-planning) Global Collaborative R&D Program explicitly emphasizes deep-dive technical discussion and joint problem-solving with Korean SMEs prior to full proposal submission. Only an iterative closed-loop collaboration (analysis → design → fabrication → test → model refinement) can realistically deliver both high-PE and high-durability targets and de-risk commercialization.
Key Words	1. Dye-based polarizing film 2. Dichroic dye 3. Polarization efficiency 4. Durability 5. PE-durability trade-off

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2) – Concept validation	End of the Project	Level (5) – Prototype demonstration
R&D Objectives	**Ultimate objective** Develop and validate core design technology for dye-based polarizing films while maintaining durability superior to iodine-based polarizers under harsh environments. **Specific objectives** 1. Identify structural factors limiting PE in dye-based polarizers (dichroic ratio/order parameter, dye concentration, film thickness) and quantify the theoretical/performance limits [5]. 2. Determine how dye–PVA interactions and fixation mechanisms affect alignment stability and PE (e.g., hydrogen bonding vs. reactive/covalent dyes) [4,6]. 3. Clarify mechanisms of PE degradation under thermal/humidity stress (alignment relaxation, dye mobility, water plasticization) and the origin of the PE–durability trade-off [3,4]. 4. Derive molecular and processing design guidelines that simultaneously satisfy high PE and high durability. 5. Fabricate and validate prototype dye-based polarizer sheets applying the design rules with the Korean SME partner.			
R&D Target Specifications	* Given the highly specific nature of the target properties, there may be no accredited testing agency available. In this case, measurements conducted internally by the Korean SME may be submitted as verification. **1) Optical performance targets** • Single-piece transmittance (Tv) ≥ 38% [7] • Polarization efficiency (PE): ≥ 98% [7] where PE=[(Tp-Tc)/(Tp+Tc)]x100%, Tp is parallel transmittance with another polarizer, and Tc is crossed transmittance with another polarizer. **2) Durability** (tests should be conducted on a polarizer laminated between two TAC films) • High-temperature exposure: 90°C for 96 h [8] |ΔPE| ≤ 2.0% no bubbling and no peeling • High-temperature / high-humidity exposure: 85°C, 85%RH for 500 h [8,9] |ΔTv| ≤ 3.0% |ΔPE| ≤ 2.0% no bubbling and no peeling 3) Technical deliverables • Prototype sheets of dye-based polarizing film meeting the above optical and durability targets • Polarizer design guidelines based on dye–PVA interactions and alignment structure • Benchmark dataset comparing optical performance and durability vs. iodine-based polarizer.			
R&D Plan	Project structure (Designated – Pre-planning) **Phase 1** (0.5 year, Pre-planning): Hypothesis verification and approach down-selection • Establish a joint evaluation protocol (Tv & PE measurement, spectral performance, and accelerated durability screening) and baseline benchmarks (iodine polarizer vs. current dye polarizer). • U of T: define multiple dye/design directions based on optical anisotropy, transition-dipole alignment, and dye–PVA interaction hypotheses; build a first-pass structure–property model for PE/Tv. • SME: fabricate a focused sample matrix reflecting U of T's design directions using practical dyeing/stretching/lamination processes; measure Tv/PE and perform short screening aging tests. • Down-select 1–2 most promising design directions with Go/No-Go criteria for Phase 2 (e.g., clear trend toward PE ≥ 97% at Tv ≥ 38% with improved durability vs. baseline). • Deliver a full Phase-2 R&D proposal (detailed work plan, resource plan, and commercialization path)			

	as required for evaluation after the pre-planning stage. **Phase 2** (2.5 years | Main R&D): Design-rule establishment and prototype demonstration • Mechanism and modeling (U of T): quantify dye–PVA interaction/fixation and its effect on alignment stability; establish the PE–durability trade-off mechanism and optimization maps linking dye structure and processing to long-term performance. • Materials/process development (Korean SMEs): implement UT design rules into industrially relevant processes (dye selection/synthesis, dyeing chemistry, stretching profile, fixation/crosslinking, lamination) and iteratively refine prototypes. • Reliability demonstration: fabricate prototype sheets and validate the optical/durability targets with additional cycling tests as needed. • Final outputs: prototype sheet(s), design guideline, and comparison dataset enabling productization for harsh-environment display markets.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	U of T's core role is to identify (at the molecular/structural level) the mechanisms that limit PE and durability in dye-based polarizers and to translate them into design rules. **Key tasks** • Analyze dye optical anisotropy (dichroic ratio, transition-dipole alignment) and quantify theoretical PE & Tv limits using established polarizer models. [5] • Model and characterize dye–PVA interactions and fixation (hydrogen bonding, electrostatic interaction, reactive/covalent dyes, crosslinking) that govern alignment stability under humidity/heat. [4,7] • Diagnose how stretching/dyeing/aging change the alignment microstructure and optical performance; clarify the origin of the PE–durability trade-off. • Provide Korean SMEs with actionable molecular/process design guidelines and parameter maps; refine them iteratively based on manufacturing/test feedback.
Expected R&R of Korean SME Partner	Korean SME will translate U of T's design rules into an industrially realizable dye-based polarizer process and validate performance and manufacturability. **Key tasks** • Select and implement dye systems and alignment/fixation conditions consistent with U of T's design directions; apply industrial processes (dyeing, stretching, lamination, etc.) to fabricate prototype polarizer films. • Verify manufacturability (process reproducibility, yield, and scale-up feasibility) and iterate prototypes based on U of T's feedback. • Perform optical characterization (Tv, PE, spectral performance) and environmental durability evaluation under the target harsh-environment conditions, including direct benchmarking vs. iodine-based polarizers. • Provide the resulting dataset to U of T for model refinement and repeated validation of the design guidelines.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	This project is expected to resolve the long-standing 'high durability vs. limited PE' challenge in dye-based polarizers by establishing a mechanism-driven, generalizable design framework. Impact • Enables next-generation polarizing films that can replace or complement iodine-based polarizers in harsh-environment display markets (automotive, outdoor, and defense), where reliability under heat/humidity/UV is critical. • Provides the Korean SME with manufacturable prototype(s), a validated design guideline, and a benchmark dataset—reducing technical risk and accelerating productization. • Demonstrates a strong global academia–SME collaboration model that connects fine-chemical (dye) innovation to high-value display materials, aligned with the TIPA Global Collaborative R&D program's commercialization and global expansion goals.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization